Practice 3

Objectives

A. Write programs that execute statements repeatedly using recursion.
B. Pass arguments by reference
C. Working with strings

Write the implementation of both functions in functions.cpp. Loops are not allowed. To compile multiple files use the following command:

```
g++ functions.cpp main.cpp
```

Problem 1

Implement the **recursive** function **isPalindrome** that returns true if the given string is a palindrome and false otherwise. The function should ignore the case of the letters. DO NOT submit the main function. The function has the following prototype:

```
// Function prototype
bool isPalindrome(const std::string& s, int left, int right);
```

You can use the following strings for testing:

```
"AbBa" is a palindrome
"Ab2Ba" is a palindrome
"AcBa" is NOT a palindrome
"" is a palindrome
"ABBA" is a palindrome
```

Problem 2

Implement the **recursive** function **findMax** that returns the maximum element int the given array. The function has the following prototype:

```
// Function prototype
int findMax(const int arr[], int size);
```